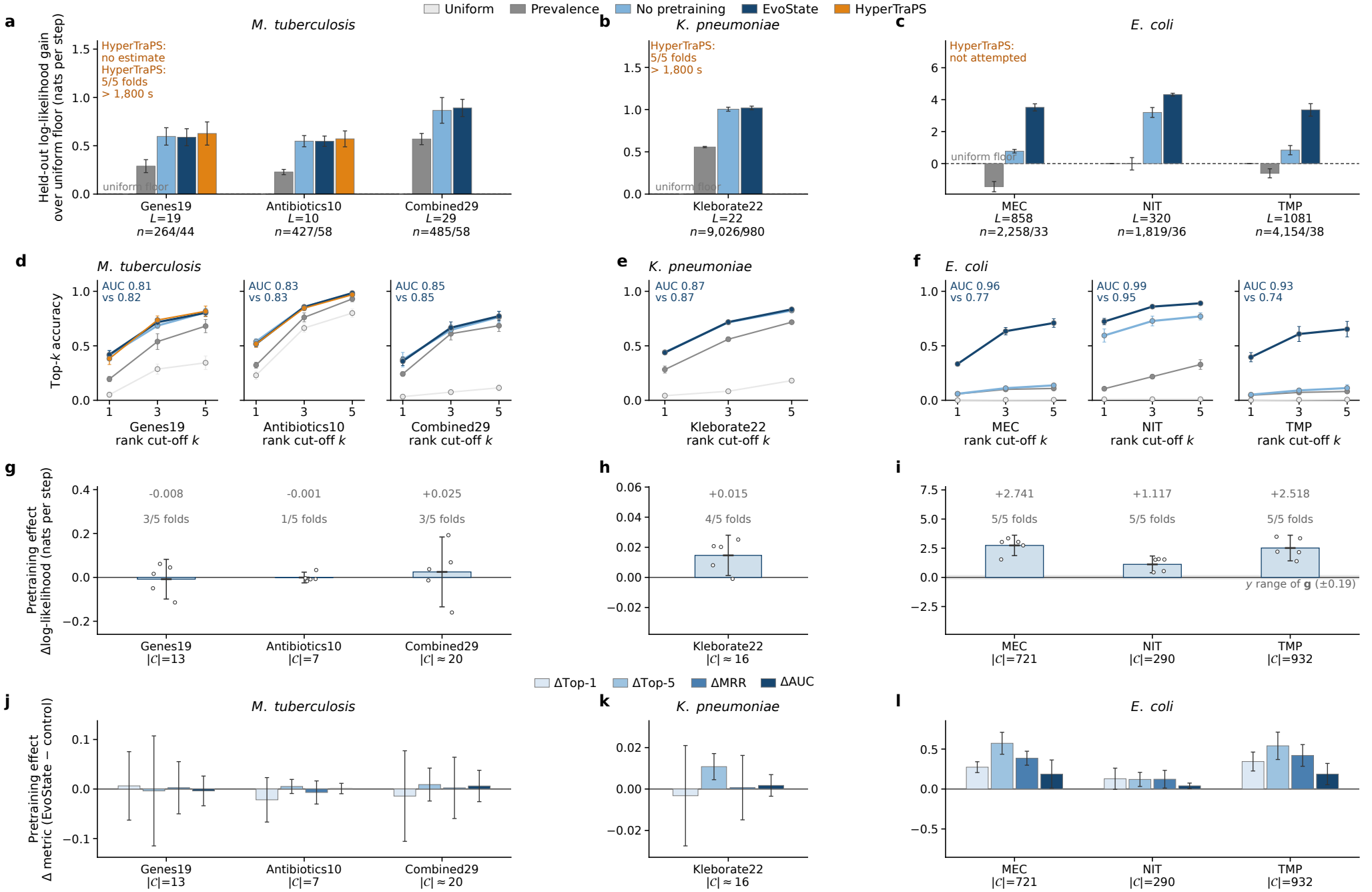

Uniform
Prevalence
No pretraining
EvoState
HyperTraPS
a
M. tuberculosis
Held-out log-likelihood gain over uniform floor (nats per step)
HyperTraPS: no estimate
HyperTraPS: 5/5 folds > 1,800 s
uniform floor
Genes19 L=19 n=264/44
Antibiotics10 L=10 n=427/58
Combined29 L=29 n=485/58
b
K. pneumoniae
HyperTraPS: 5/5 folds > 1,800 s
Kleborate22 L=22 n=9,026/980
c
E. coli
HyperTraPS: not attempted
MEC L=858 n=2,258/33
NIT L=320 n=1,819/36
TMP L=1081 n=4,154/38
d
M. tuberculosis
Top-k accuracy
AUC 0.81 vs 0.82
AUC 0.83 vs 0.83
AUC 0.85 vs 0.85
Genes19 rank cut-off k
Antibiotics10 rank cut-off k
Combined29 rank cut-off k
e
K. pneumoniae
AUC 0.87 vs 0.87
Kleborate22 rank cut-off k
f
E. coli
AUC 0.96 vs 0.77
AUC 0.99 vs 0.95
AUC 0.93 vs 0.74
MEC rank cut-off k
NIT rank cut-off k
TMP rank cut-off k
g
Pretraining effect Δlog-likelihood (nats per step)
-0.008
3/5 folds
-0.001
1/5 folds
+0.025
3/5 folds
Genes19 |C|=13
Antibiotics10 |C|=7
Combined29 |C| ≈ 20
h
+0.015
4/5 folds
Kleborate22 |C| ≈ 16
i
+2.741
5/5 folds
+1.117
5/5 folds
+2.518
5/5 folds
y range of g (±0.19)
MEC |C|=721
NIT |C|=290
TMP |C|=932
ΔTop-1
ΔTop-5
ΔMRR
ΔAUC
j
M. tuberculosis
Pretraining effect Δ metric (EvoState − control)
Genes19 |C|=13
Antibiotics10 |C|=7
Combined29 |C| ≈ 20
k
K. pneumoniae
Kleborate22 |C| ≈ 16
l
E. coli
MEC |C|=721
NIT |C|=290
TMP |C|=932